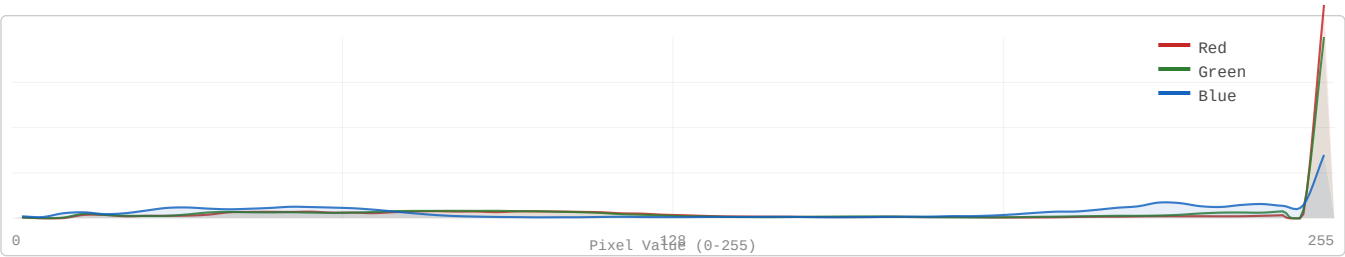
Decorrelation Gap: KODIM20

768x512 | 24-bit RGB | PNG (lossless)

Canadian Geese on Grass | Near-degenerate (PC1 = 98.42%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.9955	0.9584
G	0.9955	1.0000	0.9764
B	0.9584	0.9764	1.0000

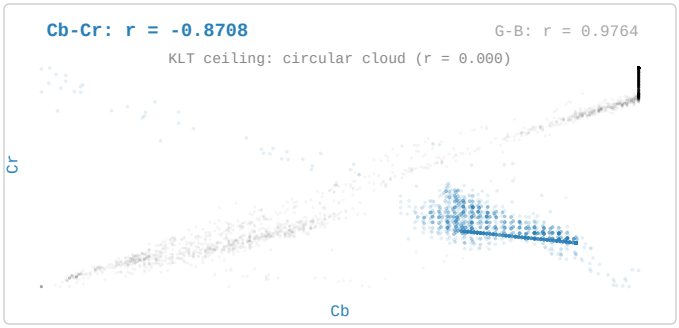
Cb-Cr: -0.871

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	0.0873	-0.1116
Cb	0.0873	1.0000	-0.8708
Cr	-0.1116	-0.8708	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.995513	0.087317	0.0873
R-B → Y-Cr	0.958368	-0.111634	0.1116
G-B → Cb-Cr	0.976444	-0.870807	0.8708
Avg |r|	0.976775	0.356586	0.3566

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

63.5% High

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

2.6%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	46.09 dB
G	49.91 dB
B	40.61 dB
Overall	43.92 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

PC1	98.42%
PC2	1.52%
PC3	0.06%

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5665	-0.5743	-0.5910	23033.47
PC2	-0.5779	-0.2345	0.7817	355.97
PC3	-0.5875	0.7843	-0.1991	13.24

8. PCA Derived Metrics

Condition Number

1739.16

PC1/PC2: 64.71 | PC2/PC3: 26.88

Blue Channel Independence

2.6%

Variance not captured by PC1

PC1 Dominant Channel

B

Loading: B (0.59)

9. Redundancy Profile

Classification: Near-degenerate | CN: 1739.16 | PC1: 98.4%

RGB Correlation: Avg |r| = 0.976775 | R-G: 0.9955 | R-B: 0.9584 | G-B: 0.9764

YCbCr Residual: Avg |r| = 0.356586 | Y-Cb: 0.0873 | Y-Cr: -0.1116 | Cb-Cr: -0.8708

Efficiency: 63.5% of KLT ceiling | Gap: 0.3566 avg |r| remaining

Blue Channel: 2.6% independent | 4:2:0 PSNR: 43.92 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.